

# No Unconditional SOTA: On the Condition-Dependent Nature of Multivariate Time Series Forecasting Models

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## ABSTRACT

Time series forecasting has seen rapid progress in recent years, with new models routinely claiming state-of-the-art (SOTA) performance. However, this paper presents systematic evidence that such SOTA claims are highly conditional on specific experimental settings—including dataset characteristics, input sequence length, and prediction horizon. Through three parallel model improvements and extensive cross-model comparisons across seven benchmark datasets, we demonstrate that no unconditional SOTA exists in long-term time series forecasting. First, we propose HYBRIDSTM, which cascades MixLinear and SimpleTM to combine time-frequency decomposition with geometric attention, achieving complementary advantages in both long-period and fine-grained scenarios. Second, we enhance SimpleTM by making its inner-product and outer-product operations learnable via softmax-weighted fusion, yielding consistent gains under independently tuned optimal configurations. Third, we design WAVEDLINEAR, which replaces the moving-average decomposition in DLinear with a learnable Stationary Wavelet Transform (SWT), achieving substantial improvements on long-sequence forecasting tasks with strong periodic patterns. Extensive experiments reveal systematic ranking reversals: SimpleTM dominates short-sequence and fine-grained scenarios, while WaveDLinear excels on long sequences with pronounced periodicity. Our ablation study across eight architectural variants further shows that the simplest WaveDLinear variant outperforms seven more complex enhancements, suggesting that additional model complexity does not guarantee better forecasting. We argue that the community should shift from pursuing unconditional SOTA toward condition-aware model selection and reporting.

## CCS CONCEPTS

• **Computing methodologies** → **Supervised learning by regression**; *Neural networks*; • **Information systems** → *Data mining*.

## KEYWORDS

Time series forecasting, wavelet transform, DLinear, SimpleTM, SOTA conditionality, linear decomposition, stationarity

## 1 INTRODUCTION

Multivariate Time Series Forecasting (MTSF) is a core problem in energy management, financial analysis, traffic planning, and numerous other domains, with deep learning achieving remarkable progress in Long-term Time Series Forecasting (LTSF). A steady stream of new models has claimed state-of-the-art (SOTA) performance [1, 3–5, 7], yet each such claim is established under a specific experimental configuration; once the dataset, input sequence length, or prediction horizon changes, the SOTA title often shifts accordingly. Moreover, multivariate time series data exhibit strong heterogeneity across variables and often suffer from limited data availability in real-world scenarios. Existing large-scale Transformer-based forecasting models typically rely on massive data and substantial computational resources, which are not always accessible, and their one-size-fits-all architectures ignore the heterogeneity among variables. These challenges motivate our exploration of alternative paradigms: starting from extremely simple yet efficient baseline models (MixLinear and SimpleTM), we aim to identify their design principles, attempt meaningful improvements, and ultimately test the hypothesis that “no unconditional SOTA exists” in time series forecasting.

Consider two representative models. DLinear [7] uses a simple moving-average decomposition followed by linear mappings and outperforms far more complex Transformer architectures on multiple benchmarks, demonstrating that “less can be more.” SimpleTM [1] further simplifies the attention mechanism by reducing self-attention to inner-product and outer-product operations, achieving leading performance on datasets such as ETTm1 with remarkably few parameters. However, our experiments reveal a systematic pattern: under short-sequence settings on ETTh1, SimpleTM significantly outperforms DLinear and its variants; yet under long-sequence settings on ETTm2, frequency-domain decomposition models decisively take the lead. The ranking of model performance undergoes systematic reversals as experimental conditions change.

This observation motivates the central question of this paper: **Does a time series forecasting model exist that unconditionally dominates all others across all settings?** If the answer is no, the community’s pursuit of unconditional SOTA warrants reconsideration.

We first systematize the comparison across five representative architectures and show that no model dominates all 42 dataset/setting combinations. Based on the observed winner patterns, we then explore three improvement directions: (1) cascading MixLinear and SimpleTM into a sequential model to combine global periodicity extraction with local detail refinement; (2) converting the fixed inner-product and outer-product operations in SimpleTM into a learnable fusion, enhancing the model’s adaptive capacity; (3) replacing the moving-average decomposition in DLinear with a learnable Stationary Wavelet Transform (SWT), inspired by MixLinear’s effective use of temporal and spectral information. Through systematic comparisons involving eight ablation variants, multiple model families, and seven benchmark datasets under diverse settings, this paper provides converging evidence that **there is no absolute SOTA model in time series forecasting—model superiority is condition-dependent**.

Our contributions are:

- We survey five representative forecasting architectures—PatchTST, iTransformer, DLinear, MixLinear, and SimpleTM—and identify the comparability problem caused by inconsistent input/output settings across original papers.
- Through a systematic 7-dataset  $\times$  6-setting comparison, we empirically demonstrate the condition-dependence of SOTA and the phenomenon of systematic ranking reversals.
- We propose HYBRIDSTM, a sequential fusion of MixLinear (time-frequency decomposition) and SimpleTM (geometric attention), designed to capture both global periodicity and local fine-grained patterns.
- We improve SimpleTM by making its inner-product/outer-product fusion learnable (termed ADASTM), motivated by the observation that inner-product and outer-product terms should have different importance across datasets.
- We design WAVEDLINEAR, which replaces the moving-average decomposition of DLinear with a learnable SWT, motivated by MixLinear’s use of time-domain and frequency-domain information.
- Through an eight-variant ablation study, we reveal the counter-intuitive finding that additional architectural complexity does not guarantee forecasting improvement—the simplest WaveDLinear variant outperforms all seven enhanced versions.

The paper follows the same logic as the accompanying presentation. Section 2 surveys the five representative models and diagnoses why their original reports are difficult to compare directly. Section 3 fixes the comparison protocol and shows the 42-setting SOTA attribution matrix. Section 4 introduces the three targeted improvements, HYBRIDSTM, ADASTM and WAVEDLINEAR. Section 5 reports the main

experimental results, while Sections 6–7 consolidate the ablation findings, practical implications, and limitations.

## 2 RELATED WORK: MODEL SURVEY AND BASELINE DIAGNOSIS

The development of time series forecasting models can be organized into three main technical trajectories: frequency/time-frequency methods, linear simplification methods, and attention simplification methods.

### 2.1 Representative Forecasting Architectures

Table 1 summarizes the five representative model families used in our systematic comparison. These models cover the dominant architectural choices in recent multivariate forecasting: patch-based temporal attention, variate-level attention, decomposition-based linear prediction, ultra-lightweight linear mixing, and lightweight signal-processing-aware attention.

This comparison also exposes a reporting problem in the literature. Although these papers often evaluate on overlapping datasets such as ETT, Electricity, Traffic, and Weather, they do not always use the same look-back windows, prediction horizons, or tuning protocols. As a result, their SOTA claims are correct only within their own experimental settings but are difficult to compare systematically across models. This motivates our 7-dataset  $\times$  6-setting benchmark, where the same model pool is evaluated under the same input/output configurations.

### 2.2 Frequency and Time-Frequency Methods

FEDformer [8] was among the early works to introduce frequency-domain information into time series forecasting. It enhances the attention mechanism via Fourier transforms, selecting dominant frequency components in the spectral domain. However, the Fourier transform is inherently global—it discards temporal locality, indicating *what* frequencies exist but not *when* they occur.

TimesNet [6] improves upon this by using FFT to discover dominant periods in the sequence and then reshaping the 1D time series into 2D tensors according to these periods for 2D convolution. While this approach uses frequency information as an auxiliary signal (for period detection), its core operations remain in the time domain.

Despite these advances, one important tool from the frequency-domain arsenal—the wavelet transform—remains underexplored in time series forecasting. Unlike the Fourier transform, the Stationary Wavelet Transform (SWT) preserves **joint time-frequency resolution**: it identifies frequency

**Table 1: Five representative time-series forecasting models and their architectural characteristics.**

| Model        | Authors / Venue                                       | Core Architecture                                                                               | Strengths                                                                                                                 | Limitations                                                                                                          |
|--------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| PatchTST     | Nie, Nguyen, Sinthong, and Kalagnanam; ICLR 2023 [5]  | Channel-independent Transformer over patched subsequences                                       | Patch tokenization reduces sequence length, preserves local temporal patterns, and works well with long look-back windows | Performance depends on patch length/stride; channel independence may underuse cross-variate structure                |
| iTransformer | Liu, Hu, Zhang, Wu, Wang, Ma, and Long; ICLR 2024 [3] | Inverted Transformer treating variates as tokens                                                | Directly models cross-variate dependencies; suitable for high-dimensional multivariate datasets such as Traffic           | Less naturally aligned with datasets where temporal periodicity dominates over variate interaction                   |
| DLinear      | Zeng, Chen, Zhang, and Xu; AAAI 2023 [7]              | Moving-average trend/seasonal decomposition plus linear heads                                   | Extremely simple, efficient, and strong on trend/periodicity-dominated settings                                           | Fixed moving-average decomposition is not adaptive and may miss multi-scale frequency structure                      |
| MixLinear    | Ma, Luo, and Sha; ICLR 2026 [4]                       | Ultra-lightweight linear mixing with low-parameter input/output mapping                         | Very small parameter count and good long-lookback efficiency; practical for resource-constrained deployment               | Limited nonlinear modeling capacity; may struggle when cross-variate interaction or fine local variation is dominant |
| SimpleTM     | Chen, Luong, Mukherjee, and Singh; ICLR 2025 [1]      | SWT/tokenization with lightweight attention based on geometric inner/outer-product interactions | Strong short-window performance; captures fine-grained local structure with few layers and parameters                     | Original inner/outer-product fusion is fixed and may not adapt to dataset-specific interaction patterns              |

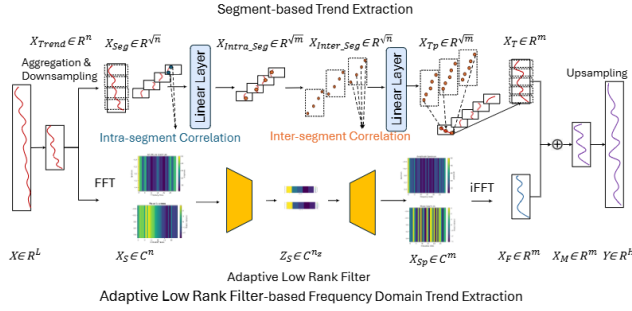
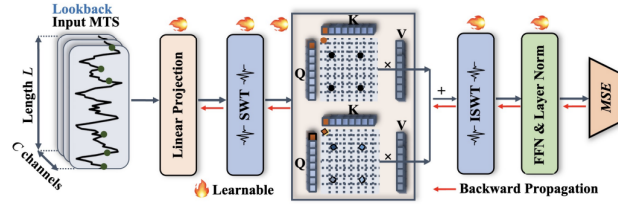


Figure 1: **MixLinear Architecture Overview.** Our dual-pathway framework efficiently processes time series data. The **Segment-based pathway** (top) downsamples input  $X \in \mathbb{R}^L$  into segments  $X_{seg} \in \mathbb{R}^{L/\pi}$ , applies linear transformations for intra-segment (blue) and inter-segment (orange) correlations, then upsamples to  $X_T \in \mathbb{R}^H$ . The **Frequency-domain pathway** (bottom) transforms segments via FFT ( $X_S \in \mathbb{C}^{L/\pi}$ ), compresses trends through adaptive low-rank filtering to latent space  $Z_S \in \mathbb{C}^{n_s}$ , reconstructs via iFFT, and outputs  $X_F \in \mathbb{R}^H$ . Final predictions  $Y \in \mathbb{R}^H$  combine both outputs, achieving competitive forecasting with only 0.1K parameters.

(a) MixLinear architecture from the original paper.



(b) SimpleTM architecture from the original paper.

**Figure 1: Original architecture figures used as the visual starting point for the improvement routes.**

components at different scales while simultaneously localizing them in time. This property is particularly valuable for real-world non-stationary time series.

## 2.3 Linear Simplification Methods

DLinear [7] represents a milestone in the linear simplification trajectory. Its minimalist architecture—moving average for trend extraction, subtraction for seasonal decomposition, and two independent linear layers for prediction—surpasses complex Transformer architectures on multiple benchmarks. This result sent a clear signal to the community: for LTSF tasks, simple linear mappings may suffice, and additional architectural complexity does not necessarily translate to commensurate gains.

RLinear [2] simplifies further by abandoning the decomposition step entirely, using only RevIN normalization with a linear mapping. Our experiments, however, suggest that the decomposition step itself is valuable—the key question is *how* to decompose. The moving average used in DLinear acts as a fixed low-pass filter capable only of coarse trend-season separation. It cannot capture multi-scale frequency structures.

## 2.4 Attention Simplification Methods

PatchTST [5] proposes segmenting time series into patches (sub-sequence fragments) before applying self-attention, effectively reducing computational complexity. iTransformer [3] inverts the Transformer’s dimensions, applying attention

along the variate dimension rather than the temporal dimension, accommodating the characteristic that in multivariate forecasting, the number of channels is typically far smaller than the sequence length.

SimpleTM [1] represents the most radical simplification along this trajectory. It reduces self-attention to two elementary operations: an inner-product term (representing query-key similarity) and an outer-product term (representing value magnitude contribution). With minimal parameters, it achieves competitive or superior performance against more complex models on multiple benchmarks. However, in the original design, the fusion of the inner and outer products is fixed—the contribution ratio of the two terms is predetermined before training and cannot adapt to different dataset characteristics and sequence properties.

### 3 UNIFIED BASELINE COMPARISON

#### 3.1 Experimental Reporting Problem

The architectural survey above exposes a key experimental problem: although PatchTST, iTransformer, DLinear, MixLinear, and SimpleTM often use overlapping benchmark datasets, their reported SOTA results are not always obtained under the same input lengths, prediction horizons, or tuning regions. One paper may emphasize short look-back windows, while another may report ultra-long look-back results. Therefore, direct comparison across original papers is not fully systematic: each model is often evaluated and highlighted in settings that match its own inductive bias.

To address this issue, we organize a unified five-model comparison over seven datasets and six input/output settings. The model set contains PatchTST, iTransformer, DLinear, MixLinear, and SimpleTM. The setting set is  $\text{seq\_len} \in \{96, 360, 720\}$  and  $\text{pred\_len} \in \{96, 192\}$ , yielding 42 dataset/setting combinations. When an exact paper-reported value is available, we use it; otherwise, we use our local reproduction under the matching setting.

#### 3.2 SOTA Attribution Across 42 Settings

**Table 2: SOTA model in each of the 42 dataset/setting combinations among PatchTST, iTransformer, DLinear, MixLinear, and SimpleTM. The winner is determined by the lowest MSE.**

| Dataset     | 96→96    | 96→192   | 360→96   | 360→192  | 720→96    | 720→192   |
|-------------|----------|----------|----------|----------|-----------|-----------|
| Electricity | SimpleTM | SimpleTM | PatchTST | SimpleTM | PatchTST  | SimpleTM  |
| ETTh1       | SimpleTM | SimpleTM | DLinear  | PatchTST | MixLinear | MixLinear |
| ETTh2       | SimpleTM | SimpleTM | DLinear  | SimpleTM | SimpleTM  | SimpleTM  |
| ETTm1       | PatchTST | SimpleTM | SimpleTM | SimpleTM | SimpleTM  | SimpleTM  |
| ETTm2       | SimpleTM | SimpleTM | SimpleTM | SimpleTM | SimpleTM  | MixLinear |
| Traffic     | PatchTST | PatchTST | SimpleTM | SimpleTM | PatchTST  | SimpleTM  |
| Weather     | SimpleTM | SimpleTM | SimpleTM | SimpleTM | SimpleTM  | SimpleTM  |

Across these 42 settings, SimpleTM is best in **30** cases, PatchTST in **7**, MixLinear in **3**, DLinear in **2**, and **none** in iTransformer. No model is optimal under all conditions. This confirms that the SOTA label is condition-dependent rather than absolute.

The winner pattern is also interpretable. SimpleTM dominates many short-window and fine-grained settings because its lightweight signal-processing-aware attention can exploit local temporal variation without excessive parameters. PatchTST becomes competitive in several longer settings because patch tokenization compresses long histories into semantically richer subseries tokens. MixLinear wins on several 720-step ETTh1/ETTm2 settings, suggesting that long histories with stable trend/periodic structure can be effectively handled by lightweight linear mixing. Although iTransformer never wins, it gets second-best on short Traffic settings, where high-dimensional cross-variate interaction is especially important. DLinear wins on ETTh1/ETTh2 at 360→96, consistent with the usefulness of explicit trend/seasonal decomposition, but its fixed moving average limits robustness across conditions.

This analysis motivates our three improvement directions. First, as MixLinear performs well at long-term, global pattern extraction and SimpleTM excels in short-term, local detail analysis, it is feasible to combine the two models' structures so as to make the most of their advantages. Second, SimpleTM's fixed inner/outer-product fusion is unlikely to be optimal for all datasets, because some settings require stronger query-key similarity modeling while others require stronger magnitude/orthogonal variation modeling. Third, MixLinear's success suggests that time-domain and frequency-domain structure should be jointly exploited, which inspires us to move beyond DLinear's fixed moving-average smoothing and replace it with a learnable SWT-based decomposition.

### 4 METHODOLOGY: PROPOSED IMPROVEMENTS

#### 4.1 Problem Formulation

Given a historical sequence  $\mathbf{X} \in \mathbb{R}^{B \times L \times C}$ , where  $B$  is the batch size,  $L$  is the input sequence length, and  $C$  is the number of variate channels, the LTSF task is to predict the next  $H$  time steps  $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times H \times C}$ . We adopt a channel-independent modeling strategy throughout.

#### 4.2 Method Overview

Based on the systematic comparison, we improve three architectural mechanisms that appear promising but still rigid. For both models, we cascade MixLinear and SimpleTM into HybridSTM, combining their characteristics of time-frequency

decomposition and geometric attention. Specifically, for SimpleTM, we make the inner-product and outer-product fusion learnable, allowing the model to adapt the relative importance of similarity matching and geometric magnitude information across datasets. And for DLinear, inspired by MixLinear’s use of both temporal and spectral information, we replace the fixed moving-average smoothing layer with a learnable SWT module, producing WAVEDLINEAR.

### 4.3 HybridSTM: Sequential Fusion of MixLinear and SimpleTM

**4.3.1 Design Motivation.** The systematic comparison in Section 3 reveals that MixLinear excels in capturing long-period global structures (e.g., on ETTh1 and ETTm2 with 720-step inputs), while SimpleTM dominates short-window and fine-grained scenarios (e.g., on ETTm1 and short ETTh1). These complementary strengths naturally suggest a sequential combination: first extract global periodic patterns and trend components using MixLinear, then refine the representation with SimpleTM’s geometry-aware attention to capture local fluctuations and details.

**4.3.2 Architectural Details.** Given an input sequence  $\mathbf{X} \in \mathbb{R}^{B \times L \times C}$ , HybridSTM processes it in two stages:

#### Stage 1: MixLinear Frontend.

- The time-domain path (TLinear) performs segment-wise linear transformation to extract local trends.
- The frequency-domain path (FLinear) uses adaptive low-rank filter to select learnable dominant frequency components, capturing global periodicities.
- The two paths are summed to produce fused features  $F_{mix} \in \mathbb{R}^{B \times L \times C}$ .

#### Stage 2: SimpleTM Backend.

- $F_{mix}$  is concatenated with the original input  $\mathbf{X}$  along the feature dimension to preserve raw fine-grained information.
- The concatenated representation is passed through SimpleTM’s SWT multi-scale tokenization and geometric attention (with inner-product and outer-product).
- A final linear layer outputs the prediction  $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times H \times C}$ .

The total parameter count is approximately the sum of MixLinear (0.1K) and SimpleTM (2.1K), i.e.,  $\approx 2.2K$ , which is only about 10% more than SimpleTM alone and remains an ultra-lightweight model.

#### 4.3.3 Expected Advantages.

- **Global-to-local modeling:** MixLinear first establishes a robust global periodic basis, then SimpleTM focuses on local refinements. This is analogous to “coarse-to-fine” multi-resolution processing.

- **Complementary inductive biases:** The combination is particularly suitable for datasets that exhibit both strong long-range periodicity and short-term non-stationary fluctuations (e.g., ETTh1, ETTm2).
- **Efficiency:** The total parameter count remains below 3K, making it suitable for resource-constrained deployment.

### 4.4 AdaSTM: Learnable Inner-Outer Product Fusion for SimpleTM

**4.4.1 Design Motivation.** SimpleTM simplifies standard self-attention to two core operations:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{InnerProduct}(\mathbf{Q}, \mathbf{K}) \cdot \mathbf{V} + \text{OuterProduct}(\mathbf{Q}, \mathbf{K}) \cdot \mathbf{V} \quad (1)$$

where the inner-product term captures alignment or similarity between tokens (i.e., whether two temporal patterns point in a similar direction in representation space), while the outer-product term captures complementary geometric magnitude information, emphasizing variation that cannot be fully explained by direct similarity alone. In the original design, these two terms are fused with a fixed equal-weight scheme. This is restrictive: in some datasets, direct similarity matching may dominate; in others, amplitude variation or orthogonal deviation may carry more forecasting signal.

**4.4.2 Learnable Fusion.** We introduce two learnable scalars  $\alpha, \beta$  that are softmax-normalized to weight the two terms:

$$\text{Score} = \frac{\exp(\alpha)}{\exp(\alpha) + \exp(\beta)} \cdot \text{Inner} + \frac{\exp(\beta)}{\exp(\alpha) + \exp(\beta)} \cdot \text{Outer} \quad (2)$$

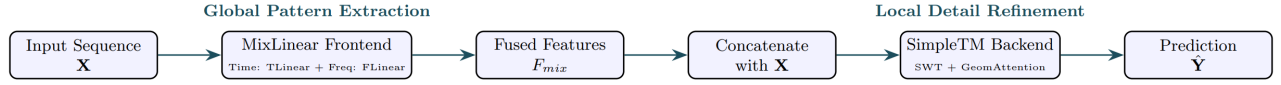
This enables the model to automatically learn the optimal balance between inner-product precision (query-key matching accuracy) and outer-product contribution (magnitude and complementary geometric variation) based on the current dataset and task characteristics. Each model is independently tuned to its optimal hyperparameter configuration, ensuring a fair comparison at each model’s best performance point.

### 4.5 WaveDLinear: SWT-Enhanced Linear Decomposition

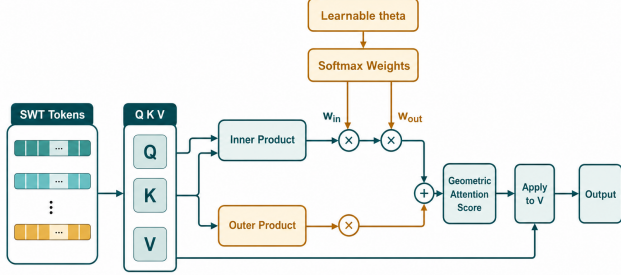
**4.5.1 Design Motivation.** The core operation of DLinear is:

$$\mathbf{X}_t = \text{AvgPool}(\mathbf{X}), \quad \mathbf{X}_s = \mathbf{X} - \mathbf{X}_t \quad (3)$$

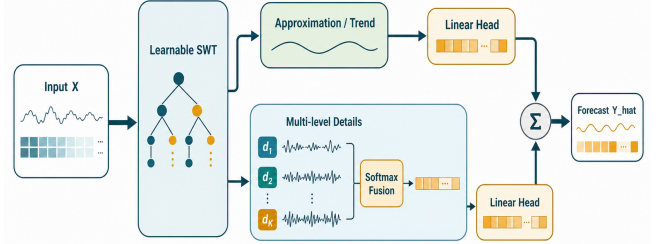
where  $\mathbf{X}_t$  is the trend component and  $\mathbf{X}_s$  is the seasonal (detail) component. AvgPool is equivalent to a fixed low-pass filter whose frequency response is determined solely by the pooling kernel size. This design has three limitations: (1) the filtering characteristics are non-learnable; (2) only single-scale coarse decomposition is possible; (3) it cannot adapt to different data properties.



(a) HYBRIDSTM: sequential fusion of MixLinear as frontend and SimpleTM as backend.



(b) ADASTM: learnable softmax fusion of inner-product and outer-product attention terms.



(c) WAVEDLINEAR: learnable SWT decomposition followed by two linear forecasting heads.

Figure 2: Architecture diagrams for the three proposed modifications.

Our core hypothesis is: *replacing the fixed low-pass filter with a learnable multi-scale frequency-domain decomposition enables the model to adaptively adjust its decomposition behavior for different datasets.*

**4.5.2 SWT Decomposition.** The Stationary Wavelet Transform is the translation-invariant variant of the discrete wavelet transform. Unlike standard Discrete Wavelet Transform (DWT), SWT does not downsample at each decomposition level, so all scale representations maintain the same temporal resolution as the original sequence. At level  $k$ , SWT decomposes the signal as:

$$\mathbf{a}^{(k)} = \mathbf{h}_{\text{low}} * \mathbf{a}^{(k-1)}, \quad \mathbf{d}^{(k)} = \mathbf{h}_{\text{high}} * \mathbf{a}^{(k-1)} \quad (4)$$

where  $\mathbf{a}^{(k)}$  is the  $k$ -th level approximation (trend) and  $\mathbf{d}^{(k)}$  is the  $k$ -th level detail (high-frequency component).  $\mathbf{h}_{\text{low}}$  and  $\mathbf{h}_{\text{high}}$  are the low-pass and high-pass filters, respectively.

**4.5.3 Key Learnable Modifications.** We introduce three critical modifications to standard SWT for the forecasting task:

**Modification 1: Learnable Filters.** The filters are changed from fixed `register_buffer` tensors to `nn.Parameter`, enabling the model to automatically learn the optimal wavelet basis shape during training. The high-pass filter is additionally constrained with a zero-mean requirement ( $\sum \mathbf{h}_{\text{high}} = 0$ ) to ensure it extracts only high-frequency components without introducing DC bias.

**Modification 2: Multi-Level Detail Adaptive Fusion.** Details at different scales contribute differently across datasets. We introduce learnable weights  $\mathbf{w} \in \mathbb{R}^K$  (where  $K$  is the number of decomposition levels), replacing equal-weight

averaging with weighted summation:

$$\mathbf{d} = \sum_{k=1}^K \mathbf{w}_k \cdot \mathbf{d}^{(k)}, \quad \mathbf{w}_k = \frac{\exp(\alpha_k)}{\sum_j \exp(\alpha_j)} \quad (5)$$

**Modification 3: Progressive Filter Normalization.** Before each forward pass, the learnable filters undergo energy normalization to prevent amplitude drift during training from degrading the decomposition quality.

The final WaveDLinear architecture is as follows: after SWT decomposition, the trend component and the fused detail component are each mapped to the prediction window via independent linear layers, and the two outputs are summed to produce the final prediction.

**4.5.4 Ablation Variants.** To systematically evaluate the contribution of different design choices, we construct eight WaveDLinear variants:

Table 3: WaveDLinear Ablation Variants.

| Variant            | Modification                                                     |
|--------------------|------------------------------------------------------------------|
| WaveDLinear (Base) | Learnable SWT + multi-level weight fusion + filter normalization |
| WaveDLinear2       | Asymmetric network: trend-linear, detail-shared MLP              |
| WaveDLinear3       | Depthwise 1D conv + GELU + Dropout on detail branch              |
| WaveDLinear4       | Per-channel soft-thresholding denoising on detail branch         |
| WaveDLinear5       | Dual-channel decoupled mixing + trend amplitude gating           |
| WaveDLinear6       | Freq-domain loss (MSE + $\alpha$ -spectral loss, $\alpha=0.1$ )  |
| WaveDLinear7       | Shape-aware loss (MSE + $\alpha$ -diff loss, $\alpha=0.2$ )      |
| WaveDLinear8       | Output-side fusion: details fused at prediction level            |

## 5 EXPERIMENTS

### 5.1 Experimental Setup

**Datasets.** We conduct experiments on seven benchmark datasets: ETTh1, ETTh2, ETTm1, ETTm2, Electricity, Traffic and Weather, with their concrete descriptions shown in Table 4.

**Table 4: All benchmark datasets adopted for experiments.**

| Name        | Description                                                    | # of dimensions |
|-------------|----------------------------------------------------------------|-----------------|
| ETTh1       | Electricity Transformer Temperature, recorded hourly           | 7               |
| ETTh2       | Electricity Transformer Temperature, recorded hourly           | 7               |
| ETTm1       | Electricity Transformer Temperature, recorded every 15 minutes | 7               |
| ETTm2       | Electricity Transformer Temperature, recorded every 15 minutes | 7               |
| Electricity | Electric power consumption                                     | 321             |
| Traffic     | Road occupancy rates                                           | 862             |
| Weather     | Meteorological indicators                                      | 21              |

**Evaluation Metrics.** Mean Squared Error (MSE) and Mean Absolute Error (MAE). Lower values indicate better performance.

**Compared Models.** We compare SimpleTM (original and improved AdaSTM), HybridSTM, PatchTST [5], iTransformer [3], MixLinear, and WaveDLinear and its eight ablation variants (WaveDLinear2–8). All experiments use a fixed random seed (2023) for reproducibility.

**Settings.** Input sequence length  $\text{seq\_len} \in \{96, 360, 720\}$ , prediction length  $\text{pred\_len} \in \{96, 192\}$ .

### 5.2 Main Results

Tables 5–11 present the full comparison across seven datasets. Best results are marked in **bold** and second-best are underlined.

**Table 5: Results on ETTh1 (MSE/MAE).**

| Seq→Pred | PatchTST             | iTransformer  | MixLinear            | SimpleTM             | HybridSTM     | AdaSTM               | WaveDLinear          |
|----------|----------------------|---------------|----------------------|----------------------|---------------|----------------------|----------------------|
| 96→96    | 0.3791/0.3954        | 0.3922/0.4088 | 0.3888/0.3936        | <u>0.3663/0.3926</u> | 0.3833/0.4032 | <b>0.3655/0.3925</b> | 0.3844/0.3927        |
| 96→192   | 0.4307/0.4313        | 0.4415/0.4349 | 0.4892/0.4670        | <u>0.4235/0.4230</u> | 0.4312/0.4337 | <b>0.4224/0.4230</b> | 0.4342/0.4212        |
| 360→96   | 0.3780/0.4022        | 0.4096/0.4209 | 0.4118/0.4180        | 0.3803/0.4079        | 0.3919/0.4125 | <u>0.3778/0.4074</u> | <b>0.3711/0.3931</b> |
| 360→192  | <u>0.4084/0.4192</u> | 0.4477/0.4459 | 0.4261/0.4263        | 0.4216/0.4325        | 0.4273/0.4382 | 0.4210/0.4329        | <b>0.4018/0.4107</b> |
| 720→96   | 0.3819/0.4125        | 0.4164/0.4416 | <u>0.3562/0.3840</u> | 0.3766/0.4151        | 0.3878/0.4137 | 0.3680/0.4083        | <b>0.3536/0.3838</b> |
| 720→192  | 0.4189/0.4356        | 0.4575/0.4679 | <b>0.4012/0.4101</b> | 0.4172/0.4392        | 0.4246/0.4462 | <u>0.4102/0.4234</u> | 0.4122/0.4348        |

#### Key observations on ETTh1:

- (1) **Short sequences ( $\text{seq\_len}=96$ ):** SimpleTM variants (SimpleTM and AdaSTM) significantly outperform all other models. AdaSTM achieves the best MSE of 0.3655 at 96→96 and 0.4224 at 96→192. WaveDLinear performs modestly in this regime.
- (2) **Long sequences ( $\text{seq\_len}=720$ ):** The situation reverses completely. WaveDLinear achieves 0.3536 MSE at 720→96, outperforming SimpleTM by 6.1%. MixLinear leads at 720→192 with 0.4012.
- (3) **Ranking reversal:** AdaSTM is SOTA at 96→96; WaveDLinear is SOTA at 720→96. *No single model is optimal across all settings.*

**Table 6: Results on ETTm1 (MSE/MAE).**

| Seq→Pred | PatchTST             | iTransformer  | MixLinear     | SimpleTM             | HybridSTM     | AdaSTM               | WaveDLinear   |
|----------|----------------------|---------------|---------------|----------------------|---------------|----------------------|---------------|
| 96→96    | <b>0.3224/0.3603</b> | 0.3366/0.3714 | 0.3870/0.3949 | <u>0.3236/0.3613</u> | 0.3306/0.3663 | 0.3280/0.3625        | 0.3491/0.3699 |
| 96→192   | <u>0.3622/0.3838</u> | 0.3808/0.3932 | 0.4175/0.4078 | <b>0.3610/0.3807</b> | 0.3724/0.3904 | 0.3673/0.3859        | 0.3886/0.3889 |
| 360→96   | <u>0.2978/0.3483</u> | 0.3038/0.3569 | 0.4083/0.4260 | <b>0.2883/0.3430</b> | 0.3080/0.3576 | 0.3001/0.3515        | 0.3050/0.3454 |
| 360→192  | 0.3440/0.3777        | 0.3404/0.3783 | 0.3547/0.3779 | <b>0.3263/0.3675</b> | 0.3531/0.3844 | <u>0.3340/0.3723</u> | 0.3372/0.3647 |
| 720→96   | <u>0.3018/0.3561</u> | 0.3173/0.3685 | 0.3670/0.3956 | <b>0.2872/0.3460</b> | 0.3120/0.3629 | 0.3031/0.3606        | 0.3102/0.3515 |
| 720→192  | 0.3394/0.3796        | 0.3500/0.3875 | 0.4352/0.4314 | <b>0.3342/0.3752</b> | 0.3466/0.3830 | 0.3458/0.3807        | 0.3397/0.3685 |

On ETTm1, SimpleTM dominates across *almost all* settings. ETTm1 has 15-minute granularity with richer high-frequency fluctuations—SimpleTM’s simplified attention mechanism appears well-suited to capturing such fine-grained temporal dependencies.

**Table 7: Results on ETTm2 (MSE/MAE).**

| Seq→Pred | PatchTST      | iTransformer  | MixLinear     | SimpleTM             | HybridSTM     | AdaSTM               | WaveDLinear          |
|----------|---------------|---------------|---------------|----------------------|---------------|----------------------|----------------------|
| 96→96    | 0.1918/0.2762 | 0.1864/0.2704 | 0.1893/0.2725 | <b>0.1739/0.2569</b> | 0.1869/0.2670 | 0.1845/0.2648        | <u>0.1821/0.2648</u> |
| 96→192   | 0.2604/0.3191 | 0.2575/0.3148 | 0.2536/0.3110 | <b>0.2387/0.2997</b> | 0.2464/0.3051 | 0.2494/0.3078        | <u>0.2463/0.3043</u> |
| 360→96   | 0.1964/0.2908 | 0.1850/0.2717 | 0.1789/0.2708 | <u>0.1680/0.2567</u> | 0.2023/0.2979 | 0.1718/0.2624        | <b>0.1640/0.2534</b> |
| 360→192  | 0.2642/0.3335 | 0.2462/0.3152 | 0.2331/0.3055 | 0.2323/0.3040        | 0.2456/0.3181 | <u>0.2297/0.2997</u> | <b>0.2190/0.2896</b> |
| 720→96   | 0.1976/0.2916 | 0.1891/0.2803 | 0.1791/0.2713 | 0.1772/0.2698        | 0.1757/0.2699 | <u>0.1709/0.2639</u> | <b>0.1614/0.2525</b> |
| 720→192  | 0.3095/0.3695 | 0.2497/0.3198 | 0.2339/0.3078 | 0.2344/0.3089        | 0.2604/0.3259 | <u>0.2310/0.3046</u> | <b>0.2153/0.2899</b> |

ETTm2 results are particularly revealing: **when  $\text{seq\_len} \geq 360$ , WaveDLinear comprehensively overtakes SimpleTM**. ETTm2 (electric load data) exhibits stronger periodic characteristics, and WaveDLinear’s SWT decomposition is precisely suited to capturing such multi-scale periodic structure.

**Table 8: Results on ETTh2 (MSE/MAE).**

| Seq→Pred | PatchTST      | iTransformer  | MixLinear     | SimpleTM             | HybridSTM            | AdaSTM               | WaveDLinear          |
|----------|---------------|---------------|---------------|----------------------|----------------------|----------------------|----------------------|
| 96→96    | 0.2928/0.3411 | 0.2974/0.3476 | 0.3071/0.3493 | <b>0.2812/0.3378</b> | 0.3013/0.3495        | <u>0.2827/0.3387</u> | 0.2904/0.3388        |
| 96→192   | 0.3750/0.3913 | 0.3809/0.3990 | 0.4082/0.4139 | <b>0.3560/0.3881</b> | <u>0.3708/0.3949</u> | <b>0.3560/0.3881</b> | 0.3747/0.3917        |
| 360→96   | 0.2828/0.3419 | 0.3264/0.3716 | 0.2923/0.3487 | 0.2824/0.3444        | 0.2909/0.3487        | <u>0.2824/0.3436</u> | <b>0.2746/0.3382</b> |
| 360→192  | 0.3609/0.3944 | 0.4152/0.4253 | 0.3515/0.3874 | <b>0.3291/0.3768</b> | 0.3578/0.3953        | <u>0.3291/0.3768</u> | 0.3356/0.3778        |
| 720→96   | 0.2795/0.3425 | 0.3103/0.3667 | 0.2930/0.3496 | 0.2769/0.3393        | 0.2995/0.3540        | <b>0.2699/0.3389</b> | <u>0.2707/0.3363</u> |
| 720→192  | 0.3525/0.3877 | 0.3708/0.4076 | 0.3478/0.3837 | <b>0.3266/0.3797</b> | 0.3747/0.3954        | <u>0.3268/0.3796</u> | 0.3333/0.3782        |

**Table 9: Results on Electricity (MSE/MAE).**

| Seq→Pred | PatchTST             | iTransformer  | MixLinear     | SimpleTM             | HybridSTM     | AdaSTM               | WaveDLinear   |
|----------|----------------------|---------------|---------------|----------------------|---------------|----------------------|---------------|
| 96→96    | 0.1753/0.2605        | 0.1721/0.2600 | 0.2098/0.2809 | <u>0.1492/0.2443</u> | 0.1623/0.2547 | <b>0.1398/0.2338</b> | 0.1973/0.2739 |
| 96→192   | 0.1818/0.2671        | 0.1834/0.2699 | 0.2054/0.2807 | <b>0.1506/0.2475</b> | 0.1740/0.2638 | <u>0.1506/0.2476</u> | 0.1971/0.2764 |
| 360→96   | <b>0.1290/0.2220</b> | 0.1365/0.2339 | 0.1456/0.2402 | 0.1305/0.2248        | 0.1474/0.2426 | 0.1357/0.2325        | 0.1395/0.2346 |
| 360→192  | 0.1470/0.2400        | 0.1556/0.2522 | 0.1592/0.2529 | 0.1424/0.2413        | 0.1510/0.2454 | <b>0.1406/0.2387</b> | 0.1536/0.2468 |
| 720→96   | <b>0.1270/0.2310</b> | 0.1368/0.2357 | 0.1389/0.2341 | 0.1309/0.2259        | 0.1409/0.2390 | 0.1313/0.2260        | 0.1330/0.2283 |
| 720→192  | 0.1430/0.2520        | 0.1555/0.2541 | 0.1542/0.2482 | <u>0.1403/0.2403</u> | 0.1528/0.2471 | <b>0.1392/0.2389</b> | 0.1480/0.2416 |

### 5.3 Eight-Model SOTA Attribution After Adding Our Improvements

We next add the three proposed improvements, HYBRIDSTM, ADASTM and WAVEDLINEAR, to the five-model attribution matrix in Table 2, resulting Table 12. To keep the comparison consistent with the literature-based five-model table, the original five models retain the same values and winners as before; only the three newly introduced models are taken

**Table 10: Results on Traffic (MSE/MAE).**

| Seq→Pred | PatchTST             | iTransformer  | MixLinear     | SimpleTM             | HybridSTM     | AdaSTM               | WaveDLinear   |
|----------|----------------------|---------------|---------------|----------------------|---------------|----------------------|---------------|
| 96→96    | <b>0.3900/0.2600</b> | 0.4046/0.2741 | 0.6628/0.3991 | 0.4200/0.2694        | 0.5015/0.3286 | 0.4213/0.2754        | 0.6461/0.3873 |
| 96→192   | <b>0.4050/0.2690</b> | 0.4215/0.2803 | 0.6109/0.3680 | 0.4345/0.2763        | 0.4785/0.3147 | 0.4342/0.2756        | 0.5989/0.3634 |
| 360→96   | 0.3620/0.2480        | 0.3950/0.2680 | 0.4144/0.2798 | <b>0.3569/0.2519</b> | 0.4329/0.3079 | <b>0.3566/0.2486</b> | 0.4092/0.2783 |
| 360→192  | 0.3780/0.2550        | 0.4070/0.2750 | 0.4272/0.2858 | <b>0.3766/0.2660</b> | 0.4031/0.2814 | <b>0.3756/0.2649</b> | 0.4214/0.2827 |
| 720→96   | <b>0.3600/0.2450</b> | 0.3910/0.2630 | 0.3894/0.2670 | 0.3609/0.2570        | 0.4422/0.3178 | <b>0.3592/0.2549</b> | 0.3872/0.2688 |
| 720→192  | 0.3750/0.2520        | 0.4020/0.2710 | 0.4035/0.2757 | <b>0.3636/0.2634</b> | 0.4010/0.2861 | <b>0.3627/0.2624</b> | 0.3977/0.2731 |

**Table 11: Results on Weather (MSE/MAE).**

| Seq→Pred | PatchTST      | iTransformer  | MixLinear     | SimpleTM             | HybridSTM            | AdaSTM               | WaveDLinear   |
|----------|---------------|---------------|---------------|----------------------|----------------------|----------------------|---------------|
| 96→96    | 0.1734/0.2167 | 0.1740/0.2240 | 0.1985/0.2375 | <b>0.1571/0.2027</b> | 0.1659/0.2094        | <b>0.1548/0.2007</b> | 0.1969/0.2367 |
| 96→192   | 0.2198/0.2570 | 0.2210/0.2650 | 0.2429/0.2728 | <b>0.2049/0.2471</b> | 0.2205/0.2593        | <b>0.2050/0.2463</b> | 0.2431/0.2739 |
| 360→96   | 0.1536/0.2043 | 0.1580/0.2110 | 0.1752/0.2269 | <b>0.1498/0.2026</b> | 0.1514/0.2011        | <b>0.1489/0.2010</b> | 0.1735/0.2245 |
| 360→192  | 0.1970/0.2441 | 0.2000/0.2510 | 0.2192/0.2635 | <b>0.1915/0.2421</b> | 0.1936/0.2455        | <b>0.1920/0.2429</b> | 0.2171/0.2611 |
| 720→96   | 0.1500/0.2005 | 0.1550/0.2080 | 0.1706/0.2256 | 0.1498/0.2043        | <b>0.1397/0.1916</b> | <b>0.1494/0.2032</b> | 0.1681/0.2219 |
| 720→192  | 0.1932/0.2416 | 0.1980/0.2490 | 0.2128/0.2608 | <b>0.1921/0.2462</b> | <b>0.1910/0.2454</b> | 0.1926/0.2440        | 0.2126/0.2622 |

from our local result logs, and a winner is replaced only when any of them obtains a lower MSE.

Under this eight-model attribution, both SimpleTM and AdaSTM are the best models in 12 settings, and they ties with each other in 2 additional settings. Then come WaveDLinear in 7, PatchTST in 5, HybridSTM in 2, MixLinear in 2, and **none** in DLinear and iTransformer. Compared with the five-model matrix, 21 out of 42 cells change—taking up a half, and every change is caused by one of the three proposed models; there is no replacement among the original five models themselves. This controlled comparison isolates the effect of our three modifications: HybridSTM replaces SimpleTM in long-window Weather settings, AdaSTM mainly replaces SimpleTM in Electricity, Traffic, Weather, and short-window ETT settings, and WaveDLinear replaces previous winners only in selected periodic or decomposition-friendly settings such as ETTh1 360→192, ETTh2 360→96, and ETTm2 720-step inputs.

#### 5.4 HybridSTM: Compound Model VS Single Model

We evaluate the effectiveness of the mixed architecture by comparing HybridSTM with its two components—MixLinear and SimpleTM—across all seven datasets under three representative input lengths: short (96→96), medium (360→96), and long (720→96). The results are reported in Table 13.

Among the 21 settings (7 datasets × 3 input lengths), SimpleTM achieves the best MSE in 18 cases, confirming its strong baseline performance, especially on short inputs. HybridSTM takes the lead in 2 settings: ETTm2 720→96 (0.1757) and Weather 720→96 (0.1397). In the latter case, HybridSTM reduces MSE by 6.7% compared to SimpleTM (0.1498) and by 18.1% compared to MixLinear (0.1706). MixLinear only wins **once**: ETTh1 720→96 (0.3562).

The fact that HybridSTM does not consistently outperform its components suggests that simply cascading two strong

models does not guarantee additive gains. The inductive biases of MixLinear (global periodicity) and SimpleTM (local detail) may interfere rather than complement each other on many datasets. However, on Weather—a dataset with both clear daily cycles and irregular fine-grained fluctuations—the sequential coarse-to-fine processing proves beneficial. This indicates that the success of hybrid architectures is highly data-dependent: under specific conditions (e.g., Weather long inputs), a well-designed fusion can surpass individual models.

#### 5.5 AdaSTM: Learnable Fusion Effect

We conduct a controlled comparison under the same forecasting setting (seq\_len=96, pred\_len=96), evaluating the original SimpleTM with fixed inner-outer product fusion against AdaSTM with learnable fusion across all seven benchmark datasets. The comparison uses matched data splits and evaluation protocol; the architectural distinction is whether the inner-outer fusion ratio is fixed before training or learned from data.

The result is striking in its consistency: **AdaSTM outperforms the fixed fusion baseline on every single dataset without exception**. While the per-dataset gain is modest (MSE reduction ranging from 0.05% to 1.9%), the seven-out-of-seven win rate indicates that learnable fusion is a robust and principled improvement over fixed fusion. Figure 3 explains why this is plausible mechanistically: ETTh1, ETTm1, Traffic, and Weather retain a high inner-product weight throughout training, whereas ETTm2 shifts toward the outer-product term and ECL steadily increases its outer-product contribution. These divergent trajectories show that the optimal fusion ratio is not a global constant; it depends on the temporal structure of the dataset.

#### 5.6 Ablation Study: WaveDLinear Variants

To isolate the contribution of each design choice, we conduct a systematic ablation of WaveDLinear variants. Table 15 presents the key results on ETTh1 at seq\_len=720 with db2 wavelet initialization.

Three key findings emerge from the ablation:

- (1) **SWT substitution is the primary gain source.** Simply replacing the moving average with a fixed SWT (Stage 1) reduces MSE from 0.4018 to 0.3750 (6.6% improvement), confirming that frequency-domain decomposition is inherently better suited to the task than a fixed low-pass filter.
- (2) **Learnability provides further gains.** Converting filters from fixed to learnable (Stage 2) further reduces the 96-step MSE to 0.3675 (8.5% improvement over DLinear). The gain from learnable weights (Stage 3) is modest on this particular configuration, while filter

**Table 12: SOTA attribution after adding HYBRIDSTM, ADASTM and WAVEDLINEAR. Changes from the five-model table occur only when one of the three new models exceeds the previous winner.**

| Dataset     | 96→96         | 96→192          | 360→96             | 360→192            | 720→96             | 720→192            |
|-------------|---------------|-----------------|--------------------|--------------------|--------------------|--------------------|
| Electricity | <b>AdaSTM</b> | SimpleTM        | PatchTST           | <b>AdaSTM</b>      | PatchTST           | <b>AdaSTM</b>      |
| ETTh1       | <b>AdaSTM</b> | <b>AdaSTM</b>   | <b>WaveDLinear</b> | <b>WaveDLinear</b> | <b>WaveDLinear</b> | MixLinear          |
| ETTh2       | SimpleTM      | SimpleTM/AdaSTM | <b>WaveDLinear</b> | SimpleTM/AdaSTM    | <b>AdaSTM</b>      | SimpleTM           |
| ETTh1       | PatchTST      | SimpleTM        | SimpleTM           | SimpleTM           | SimpleTM           | SimpleTM           |
| ETTh2       | SimpleTM      | SimpleTM        | <b>WaveDLinear</b> | <b>WaveDLinear</b> | <b>WaveDLinear</b> | <b>WaveDLinear</b> |
| Traffic     | PatchTST      | PatchTST        | <b>AdaSTM</b>      | <b>AdaSTM</b>      | <b>AdaSTM</b>      | <b>AdaSTM</b>      |
| Weather     | <b>AdaSTM</b> | SimpleTM        | <b>AdaSTM</b>      | SimpleTM           | <b>HybridSTM</b>   | <b>HybridSTM</b>   |

**Table 13: Comparison of HybridSTM, MixLinear and SimpleTM under different input lengths (seq\_len=96/360/720→96, MSE). Best results are in bold.**

| Dataset     | 96→96     |               |           | 360→96    |               |           | 720→96        |               |               |
|-------------|-----------|---------------|-----------|-----------|---------------|-----------|---------------|---------------|---------------|
|             | MixLinear | SimpleTM      | HybridSTM | MixLinear | SimpleTM      | HybridSTM | MixLinear     | SimpleTM      | HybridSTM     |
| Electricity | 0.2098    | <b>0.1492</b> | 0.1623    | 0.1456    | <b>0.1305</b> | 0.1474    | 0.1389        | <b>0.1309</b> | 0.1409        |
| ETTh1       | 0.3888    | <b>0.3663</b> | 0.3833    | 0.4118    | <b>0.3803</b> | 0.3919    | <b>0.3562</b> | 0.3766        | 0.3878        |
| ETTh2       | 0.3071    | <b>0.2812</b> | 0.3013    | 0.2923    | <b>0.2824</b> | 0.2909    | 0.2930        | <b>0.2769</b> | 0.2995        |
| ETTh1       | 0.3870    | <b>0.3236</b> | 0.3306    | 0.4083    | <b>0.2883</b> | 0.3080    | 0.3670        | <b>0.2872</b> | 0.3120        |
| ETTh2       | 0.1893    | <b>0.1739</b> | 0.1869    | 0.1789    | <b>0.1680</b> | 0.2023    | 0.1791        | 0.1772        | <b>0.1757</b> |
| Traffic     | 0.6628    | <b>0.4200</b> | 0.5015    | 0.4144    | <b>0.3569</b> | 0.4329    | 0.3894        | <b>0.3609</b> | 0.4422        |
| Weather     | 0.1985    | <b>0.1571</b> | 0.1659    | 0.1752    | <b>0.1498</b> | 0.1514    | 0.1706        | 0.1498        | <b>0.1397</b> |

**Table 14: AdaSTM vs. SimpleTM across seven datasets (seq\_len=96→96, MSE). Lower is better. Each model is independently tuned to its optimal hyperparameter configuration.**

| Dataset     | SimpleTM (fixed) | AdaSTM (learnable) | $\Delta$ MSE |
|-------------|------------------|--------------------|--------------|
| ETTh1       | 0.3706           | <b>0.3701</b>      | -0.0005      |
| ETTh2       | 0.2850           | <b>0.2797</b>      | -0.0053      |
| ETTh1       | 0.3208           | <b>0.3202</b>      | -0.0006      |
| ETTh2       | 0.1798           | <b>0.1783</b>      | -0.0016      |
| Electricity | 0.1682           | <b>0.1660</b>      | -0.0021      |
| Traffic     | 0.4642           | <b>0.4630</b>      | -0.0012      |
| Weather     | 0.1708           | <b>0.1695</b>      | -0.0013      |

normalization with the zero-mean constraint (Stage 4) slightly degrades performance, suggesting the constraints may over-regularize the learned filters.

- (3) **Complexity does not bring gains.** Seven enhanced variants (Stages 5–11) *none* surpass the simplest version. Adding asymmetric networks, depthwise convolutions, soft-thresholding, frequency-domain constraints, shape-aware constraints, or output-side fusion all fail to improve performance—some even cause degradation (WaveDLinear6: 0.3913, worse than Stage

**Table 15: Ablation study on ETTh1 (seq\_len=720). DLinear baseline: MSE 0.4018 (96-step), 0.4312 (192-step).**

| Stage | Description                        | 96-step MSE | 192-step MSE |
|-------|------------------------------------|-------------|--------------|
| —     | DLinear (baseline)                 | ~0.4018     | ~0.4312      |
| 1     | + Fixed SWT + Mean-Centering       | 0.3750      | 0.4131       |
| 2     | + Learnable filters (nn.Parameter) | 0.3675      | 0.4022       |
| 3     | + Learnable detail weights         | 0.3675      | 0.4022       |
| 4     | + Filter norm + zero-mean cstr.    | 0.3701      | 0.4076       |
| 5     | WaveDLinear2: asymmetric MLP       | 0.3832      | 0.4372       |
| 6     | WaveDLinear3: depthwise conv       | 0.3847      | 0.4346       |
| 7     | WaveDLinear4: soft-threshold       | 0.3851      | 0.4337       |
| 8     | WaveDLinear5: dual-channel mix     | 0.3870      | 0.4369       |
| 9     | WaveDLinear6: freq-domain loss     | 0.3913      | 0.4416       |
| 10    | WaveDLinear7: shape-aware loss     | 0.3847      | 0.4342       |
| 11    | WaveDLinear8: output-side fusion   | 0.3940      | 0.4368       |

- 2). This counter-intuitive finding aligns with DLinear’s original observation that “simple linear models can outperform complex Transformers.”

## 5.7 Extended Results: WaveDLinear Variants on ETTh1

For completeness, Table 16 presents the full WaveDLinear variant comparison across all ETTh1 sequence settings.

The results reinforce our earlier finding: the simplest WaveDLinear (Base) achieves the best performance at 720→96 (0.3536 MSE), while no single variant dominates all settings. Interestingly, WaveDLinear8 (output-side fusion) achieves the best result at 720→192, suggesting that the optimal fusion strategy may itself be condition-dependent.

## 6 DISCUSSION

### 6.1 Summary of Key Findings

Our experiments reveal three interrelated findings:

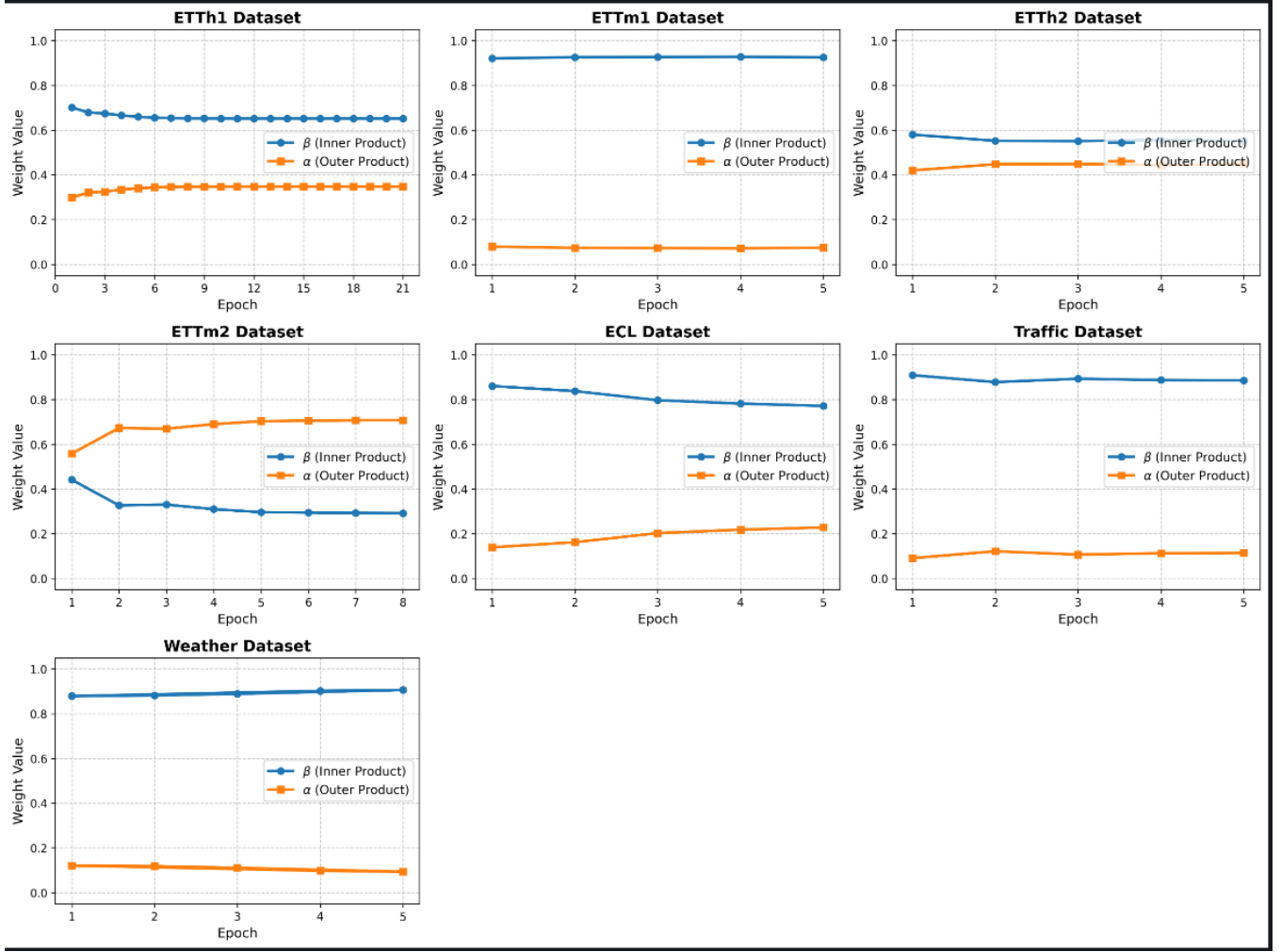


Figure 3: Training dynamics of AdaSTM’s learnable fusion weights across seven datasets. The blue curve denotes the weight assigned to the inner-product term ( $\beta$ ), and the orange curve denotes the weight assigned to the outer-product term ( $\alpha$ ). Different datasets converge to visibly different inner–outer product balances, supporting the need for data-adaptive fusion rather than a single fixed ratio.

Table 16: Full WaveDLinear variant comparison on ETTh1 (MSE/MAE). Best in bold, second-best underlined.

| Setting | Base                 | WaveD-2              | WaveD-3       | WaveD-4              | WaveD-5       | WaveD-6       | WaveD-7              | WaveD-8              |
|---------|----------------------|----------------------|---------------|----------------------|---------------|---------------|----------------------|----------------------|
| 96→96   | 0.3844/0.3927        | <b>0.3832/0.3920</b> | 0.3847/0.3929 | 0.3851/0.3928        | 0.3870/0.3949 | 0.3913/0.3981 | 0.3847/0.3927        | 0.3940/0.4005        |
| 96→192  | <u>0.4342/0.4212</u> | 0.4372/0.4230        | 0.4346/0.4210 | <b>0.4337/0.4192</b> | 0.4369/0.4221 | 0.4416/0.4271 | <u>0.4342/0.4212</u> | 0.4368/0.4224        |
| 360→96  | 0.3711/0.3931        | <b>0.3689/0.3914</b> | 0.3743/0.3968 | <u>0.3689/0.3925</u> | 0.3697/0.3955 | 0.3770/0.3981 | 0.3707/0.3928        | 0.3696/0.3917        |
| 360→192 | 0.4018/0.4107        | 0.4032/0.4115        | 0.4109/0.4207 | <u>0.4035/0.4141</u> | 0.4037/0.4153 | 0.4092/0.4173 | <b>0.4012/0.4103</b> | 0.4029/0.4120        |
| 720→96  | <b>0.3536/0.3838</b> | 0.4083/0.4277        | 0.3800/0.4031 | <u>0.3733/0.3974</u> | 0.3777/0.4035 | 0.3849/0.4076 | 0.4107/0.4281        | 0.3747/0.3977        |
| 720→192 | <u>0.4102/0.4234</u> | 0.4521/0.4513        | 0.4266/0.4425 | <u>0.4166/0.4343</u> | 0.4162/0.4265 | 0.4209/0.4286 | 0.4177/0.4332        | <b>0.4099/0.4207</b> |

(1) **SOTA condition-dependence:** The ranking of model performance undergoes systematic reversals depending on dataset characteristics and input sequence length.

After adding HybridSTM, AdaSTM and WaveDLinear to the fixed five-model baseline table, AdaSTM is best

in 12 settings, SimpleTM remains best in 12, WaveDLinear is best in 7, and HybridSTM only wins in 2 settings. The gains of the new models are therefore real but condition-specific, and no model is unconditionally optimal.

- (2) **The effectiveness of simplicity:** The simplest WaveDLinear variant—only replacing the moving average with a learnable SWT—outperforms all seven enhanced variants. Every attempt to increase complexity (asymmetric architectures, convolutional enhancement, frequency-domain losses, shape constraints, output-side fusion) failed to yield consistent improvements. This principle is also testified in HybridSTM: Though the mixed structure seems one-fits-all, it seldom guarantees higher accuracy.
- (3) **The value of learnability:** Both the inner-outer product fusion learnability in AdaSTM and the SWT filter learnability in WaveDLinear yield stable, consistent gains. However, the improvements are modest—indicating that the choice of base architecture (temporal attention vs. frequency-domain decomposition) has a greater impact on performance than fine-tuning within a given architecture.

## 6.2 Engagement with Prior Work

Our results validate DLinear’s core claim—“simple designs can surpass complex Transformers”—but extend it further: this advantage is itself condition-dependent. DLinear-type models (including WaveDLinear) perform best under strong periodicity and long input sequences but fall short under short-sequence, fine-grained scenarios where SimpleTM excels.

For SimpleTM, we confirm the effectiveness of its minimalist attention design while identifying learnable inner-outer product fusion as a valuable improvement direction. More importantly, we demonstrate that even among the simplest model designs, no single architectural choice dominates across all scenarios.

These findings contrast with frequency-domain methods like FEDformer and TimesNet: our SWT approach introduces frequency information while preserving temporal locality—an advantage that materializes specifically on strongly periodic datasets like ETTh2. This does not imply that SWT is categorically superior to the Fourier transform; rather, it demonstrates that under specific data conditions, the benefit of joint time-frequency resolution can be realized.

## 6.3 Practical Implications

For time series forecasting practitioners, the core recommendation from this work is: **select models based on data**

**characteristics, not the latest SOTA ranking.** Specifically:

- **Long sequences + strong periodicity** (e.g., electric load): frequency-aware linear/decomposition models such as WaveDLinear or MixLinear can be preferable, but the exact winner still depends on the dataset and horizon.
- **Short sequences + fine-grained data** (e.g., high-frequency sensors): simplified attention models, especially SimpleTM/AdaSTM, are more suitable.
- **Large-scale deployment:** linear simplification models have smaller parameter counts and faster inference, offering greater practical value—the SOTA models reported in papers are research benchmarks rather than direct production solutions.

## 6.4 Limitations

This work has several limitations:

- (1) **Dataset coverage:** While we cover seven benchmark datasets, several commonly used forecasting benchmarks (e.g., Exchange, ILI) are not included. Whether the SOTA reversal patterns generalize consistently to these datasets remains to be verified.
- (2) **SWT limitations:** Wavelet transforms excel at time-frequency localization, but for purely global periodic signals, the Fourier transform has advantages, since FFT can precisely locate frequencies, while SWT’s frequency resolution is constrained by the number of decomposition levels.
- (3) **AdaSTM gain magnitude:** While learnable fusion achieves a 7/7 win rate across all datasets, per-dataset gains are moderate (MSE reduction ranging from 0.1% on ETTh1 to 1.9% on ETTh2). Whether tuning hyperparameters specifically for AdaSTM can amplify these gains remains to be investigated.
- (4) **Time complexity:** Although a comprehensive examination is conducted across 6 length settings, 7 datasets and 8 models, we merely pay attention to their accuracy by comparing their MSE and MAE, ignoring the time consumption under different circumstances. Specifically, the SWT-and-Transformer-based SimpleTM usually take quite a long training time. Parallel filters mentioned by its paper [1] might be a feasible route for efficiency optimization.
- (5) **No adaptive selection strategy:** While we identify the phenomenon of “SOTA conditionality,” this paper does not provide a solution for automatically selecting the optimal model for a given dataset.

## 6.5 Future Directions

Based on our findings, we propose the following future research directions:

- **Larger model  $\times$  condition matrices:** Since systematic comparisons across a broader set of datasets and more fine-grained conditions (different channel counts, noise levels, missing-data rates) are conducted, can we establish a more comprehensive “model-condition compatibility table,” serving as a practical reference for the community?
- **Condition-aware model selection:** Can we design an algorithm-powered meta-learner that automatically recommends the most suitable model architecture based on statistical properties of the dataset (periodicity strength, autocorrelation structure, noise levels, etc.)?
- **Adaptive fusion architectures:** If different models dominate under different conditions, can we design a dynamic routing mechanism that allows the model to adaptively switch between or combine different prediction strategies based on the input?

## 7 CONCLUSION

This paper explores the improvement of time series forecasting models from three directions—cascading MixLinear and SimpleTM, adjusting SimpleTM’s inner-outer product operations with learnable fusion, and enhancing DLinear’s decomposition layer with a learnable Stationary Wavelet Transform—and through systematic multi-model, multi-condition comparisons across multiple benchmark datasets, demonstrates that no unconditional SOTA exists in time series forecasting.

WaveDLinear achieves an 8.5% MSE improvement over DLinear on ETTh1 long sequences ( $0.4018 \rightarrow 0.3675$ ) by replacing the moving average decomposition with a learnable SWT, and it replaces previous five-model winners in four of the 42 unified settings. AdaSTM achieves a 7/7 win rate against fixed-fusion SimpleTM in the controlled fusion study and becomes the single best model in 12 settings in the eight-model attribution matrix. Ablation results across eight WaveDLinear variants demonstrate that the simplest version outperforms all more complex enhancements, reinforcing the “simplicity as a virtue” design philosophy.

However, the most important contribution of this work is not a model but an observation: SimpleTM is stronger on short-sequence and fine-grained scenarios; WaveDLinear excels on long-sequence, strongly periodic data; PatchTST and iTransformer each have their zones of advantage—**SOTA is a condition-dependent, relative concept, not an absolute label**. The community should shift from chasing “which model is best under a specific setting” to understanding “which model is suitable for which conditions.”

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## APPENDIX

### A MODEL NAMING CONVENTION

The model naming conventions used in this paper are as follows:

- **HybridSTM:** A sequential model where MixLinear (time-frequency decomposition) serves as the frontend and SimpleTM (SWT + geometric attention) serves as the backend. The two stages are connected via concatenation. Total parameters  $\approx 2.2K$ .
- **AdaSTM (Adaptive SimpleTM):** SimpleTM with its fixed inner-product/outer-product fusion replaced by learnable softmax-normalized weight pairs (`io_mix_logits`, `nn.Parameter`).
- **WaveDLinear (Base):** DLinear with learnable Stationary Wavelet Transform (SWT) replacing the moving-average decomposition. Three core modifications: learnable wavelet filters (`nn.Parameter`), multi-level detail adaptive fusion weights, and progressive filter normalization with zero-mean constraints.
- **WaveDLinear2–8:** Enhanced variants built on top of WaveDLinear (detailed in Section 4.5.4). None consistently surpass the base version.

## B REPRODUCIBILITY

All experiments use a unified random seed (2023) to ensure reproducibility. Complete experimental logs and raw output files are organized under the `MixLinear/` and `SimpleTM/`

directories. The SWT decomposition uses db2 (Daubechies 2) wavelet initialization by default; Haar and random initializations are also supported. All models are implemented in PyTorch and trained on NVIDIA GPUs.